SG No. 46 C_{2v}^{22} $Ima2$ [orthorhombic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$
* Wyckoff site: `4a`, site symmetry: `..2`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[0,\ 0,\ z]$	`[1,-2]`
2	$[X,\ -Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[3,-4]`

Table 2: Wyckoff bond: `4b@4a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	`[1,2]`
2	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[3,4]`

Table 3: Wyckoff bond: `8c@4a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ z]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$[0,\ 0,\ z]$	`[2]`
3	$[X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[3]`
4	$[-X,\ Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[4]`

* Wyckoff site: `4b`, site symmetry: `m..`

Table 4: Wyckoff bond: `4a@4b`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0,\ Y,\ Z]$	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,4]`
2	$[0,\ -Y,\ Z]$	$\left[\frac{3}{4},\ -y,\ z\right]$	`[2,3]`

Table 5: Wyckoff bond: `4b@4b`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,-4]`
2	$[-X,\ 0,\ 0]$	$\left[\frac{3}{4},\ -y,\ z\right]$	`[2,-3]`

Table 6: Wyckoff bond: `8c@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{4},\ y,\ z\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{3}{4},\ -y,\ z\right]$	`[2]`
3	$[X,\ -Y,\ Z]$	$\left[\frac{3}{4},\ -y,\ z\right]$	`[3]`

4	$[-X, Y, Z]$	$\left[\frac{1}{4}, y, z\right]$	[4]

* Wyckoff site: 8c, site symmetry: 1

Table 7: Wyckoff bond: 8a@8c, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$[-x, -y, z]$	[2]
3	$[X, -Y, Z]$	$\left[x + \frac{1}{2}, -y, z\right]$	[3]
4	$[-X, Y, Z]$	$\left[\frac{1}{2} - x, y, z\right]$	[4]